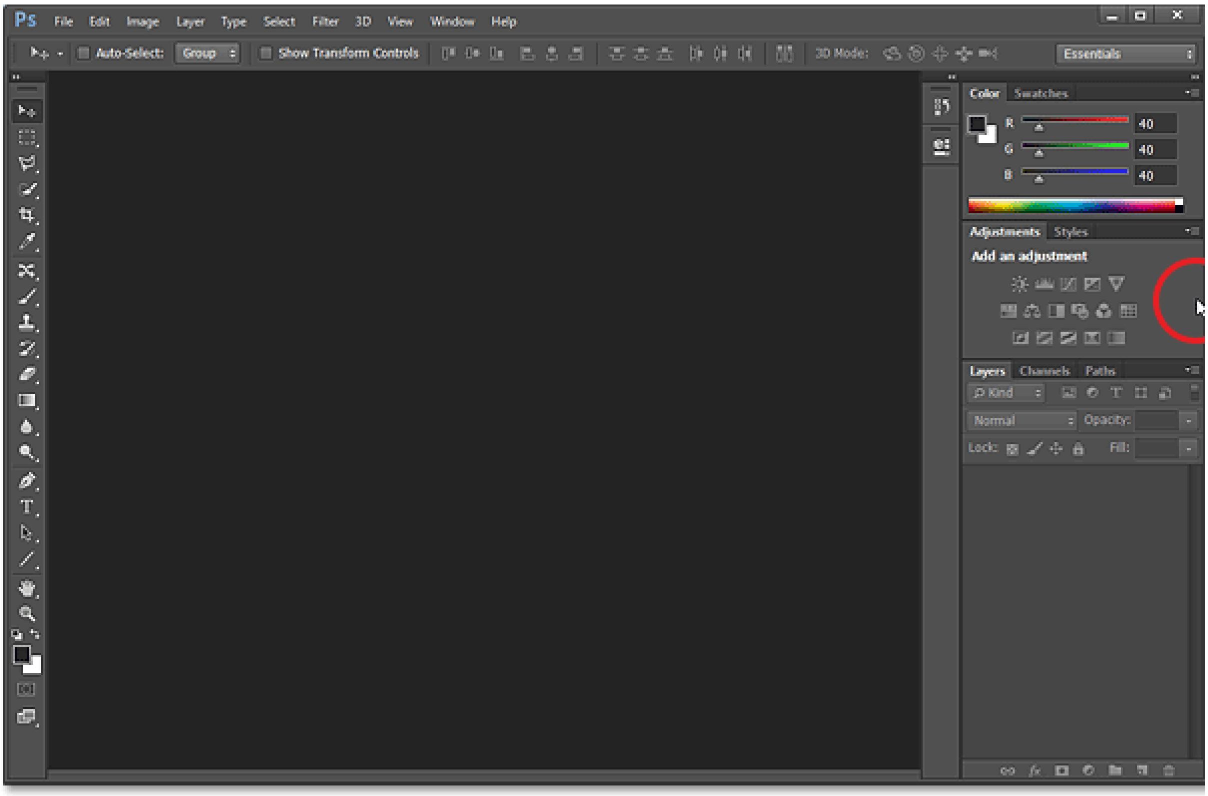
Unit:1 Introduction to Image Editing

- Working with Interface
- Resolution and Image Size
- Color Spaces and Color Modes
- Brief Overview of Tools
- Selection tools, Filling stroking tools
- Blending options
- Content Adjustment option
- History option.

Working with Interface

The elements of Photoshop Interface can be divided in four parts.

1. Menu bar
2. Toolbox
3. Control Panel (Option bar)
4. Panes (Panel)



1. **Menu bar:**

   Menus are probably the most **familiar interface elements to a new Photoshop user**. They contain all sorts of options. The menu items allows access different commands to use in Photoshop. For example, from the File menu, you can open and save files. The Image menu allows you to make various adjustments, such as the image size, whereas the **Filter menu gives you access to more advanced tools and effects.** The various menus of photoshops are File, Edit, Image, Select, Layer, Filter, Windows and Helps.

2. **Toolbox:**

   This is one of the most important features in Photoshop—you can select different tools for **editing your images**. It contains a bunch of icons that represent the different tools Photoshop offers to alter and create images. These include tools for selecting specific areas of images, **changing the colors of the image, stretching, transforming, and erasing parts of an image, and many more.** To get an idea of what some of these tools can do, mouse over the icons and you'll get an explanatory **tool tip.**

3. **Option bar:**

   The options bar, which is located directly under the menus, is a useful tool when working with the different Photoshop tools. **It shows different options available with various tools**. When you change the tool from toolbox it will shows you different options available to particular tools.

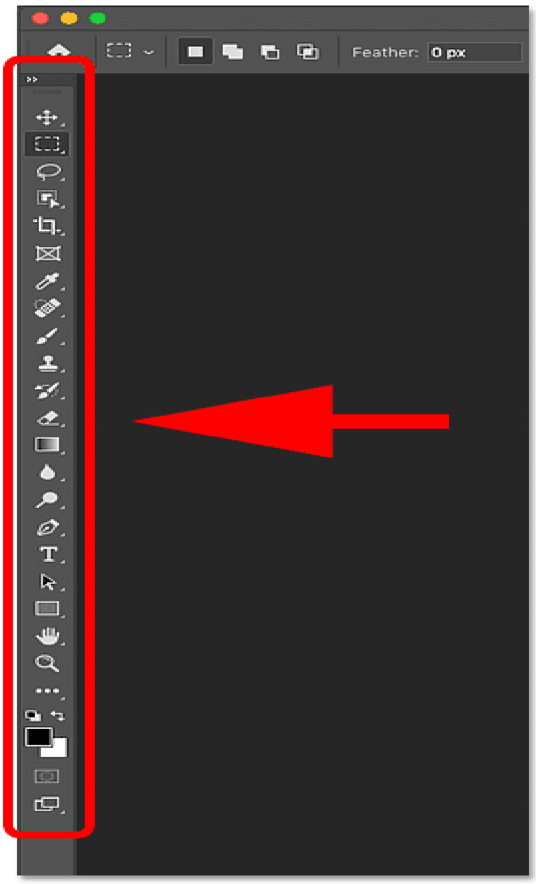
4. **Panels:**

   Panels are also important features of the Photoshop interface. All sorts of information are displayed in these panes. **They display location information, tool options, and history, among other things. You can display and hide various panels from Window menu of menu bar.**

Brief Overview of Tools

The Photoshop toolbar

Photoshop's toolbar is located along the left of the screen:



Choosing a single or double column toolbar

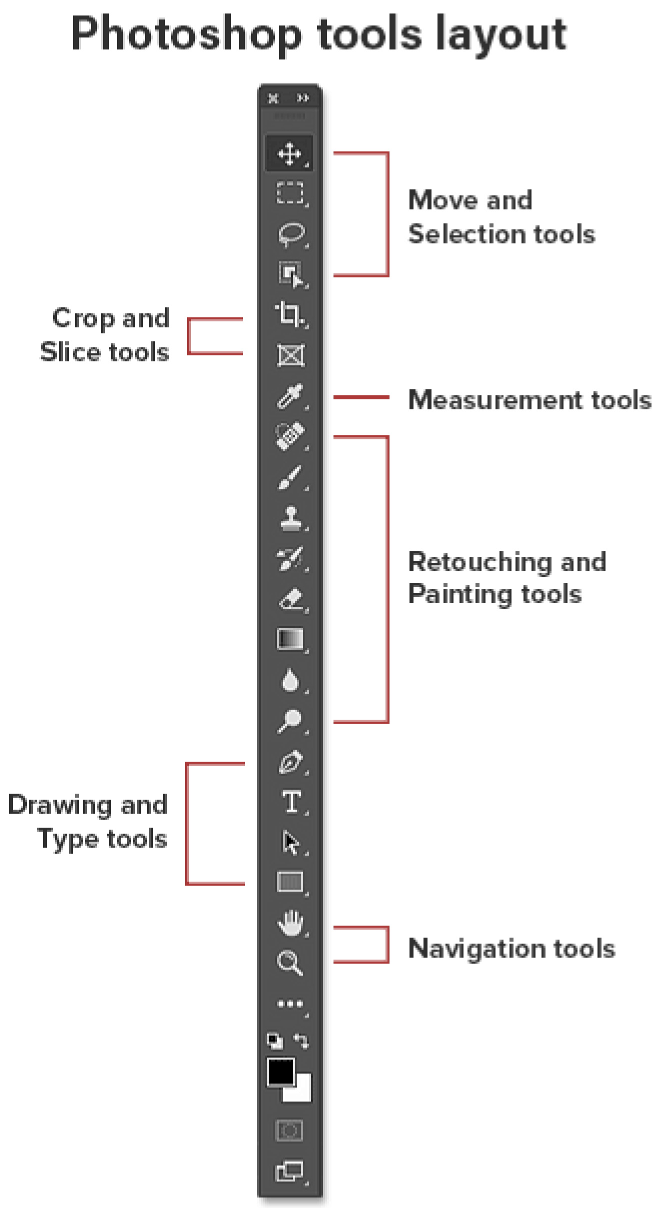
By default, the toolbar appears as a long, single column. But it can be expanded into a shorter, double column by clicking the **double arrows** at the top. Click the double arrows again to return to a single column toolbar:

The tools layout

Let's look at how Photoshop's toolbar is organized. While it may seem like the tools are listed randomly, there's actually a logical order to it, with related tools grouped together.

At the top, we have Photoshop's **Move** and **Selection** tools. And directly below them are the **Crop** and **Slice** tools. Below that are the **Measurement** tools, followed by Photoshop's many **Retouching** and **Painting** tools.

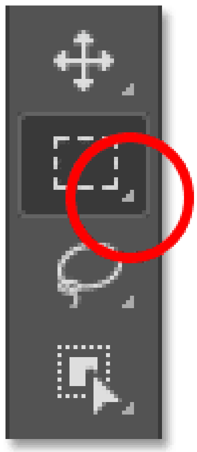
Next are the **Drawing** and **Type** tools. And finally, we have the **Navigation** tools at the bottom:



The toolbar's hidden tools

Each tool in the toolbar is represented by an icon, and there are many more tools available than what we see.

A small **arrow** in the bottom right corner of a tool icon means that there are more tools hiding behind it in that same spot:

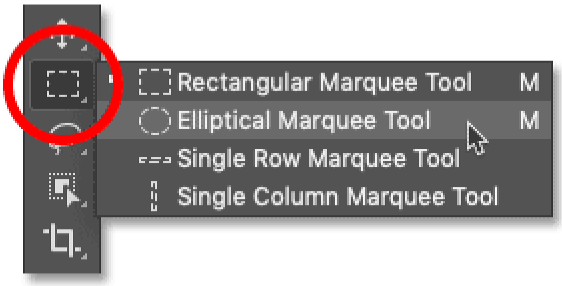


Most of the spots in the toolbar hold more than one tool.

To view the additional tools, **click and hold** on the icon. Or **right-click** (Win) / **Control-click** (Mac) on the icon. A fly-out menu will open listing the other tools that are available.

For example, if I click and hold on the Rectangular Marquee Tool icon, the fly-out menu tells me that along with that tool, the Elliptical Marquee Tool, the Single Row Marquee Tool and the Single Column Marquee Tool are also grouped in with it.

To choose one of the additional tools, click on its name in the list. I'll choose the Elliptical Marquee Tool:



Choosing a hidden tool from the fly-out menu.

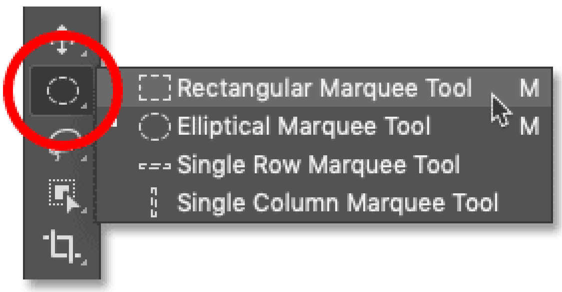
The default tool

The tool that's initially displayed in each spot in the toolbar is known as the **default tool**. For example, the Rectangular Marquee Tool is the default tool for the second spot from the top. But Photoshop won't always display the default tool. Instead, it will display the last tool you selected.

Notice that after choosing the Elliptical Marquee Tool from the fly-out menu, the Rectangular Marquee Tool is no longer displayed in the toolbar. The Elliptical Marquee Tool has taken its place:

Each spot in the toolbar displays either the default tool or the last tool selected.

To select the Rectangular Marquee Tool at this point, I would need to either **click and hold**, or **right-click** (Win) / **Control-click** (Mac), on the Elliptical Marquee Tool icon. Then I could select the Rectangular Marquee Tool from the menu:



Selecting the Rectangular Marquee Tool from behind the Elliptical Marquee Tool.

A summary of Photoshop's tools

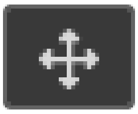
So now that we've learned how Photoshop's toolbar is organized, let's look at the tools themselves.

Below is a quick summary of each of Photoshop's tools, along with a brief description of what each tool is used for. The tools are listed in order from top to bottom, and specific tools are covered in more detail in other lessons.

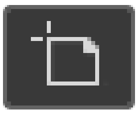
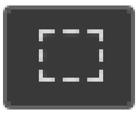
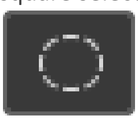
An asterisk (*) after a tool's name indicates a default tool, and the letter in parenthesis is the tool's keyboard shortcut. To cycle through tools with the same keyboard shortcut, press and hold **Shift** as you press the letter.

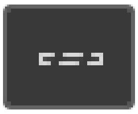
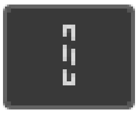
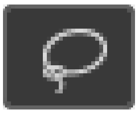
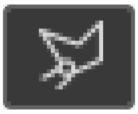
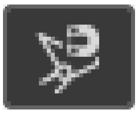
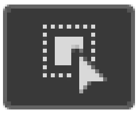
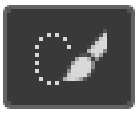
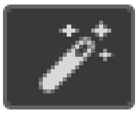
This list is up-to-date as of Photoshop 2023. Note that some tools are not available in earlier versions.

- ### Move and Selection tools

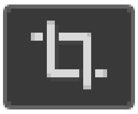
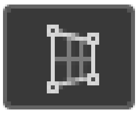
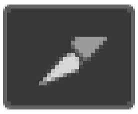
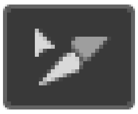
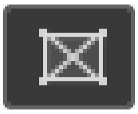
  

  Move Tool * (V): The Move Tool is used to move layers, selections and guides within a Photoshop document. Enable "Auto-Select" to automatically select the layer or group you click on.

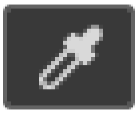
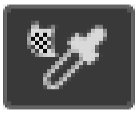
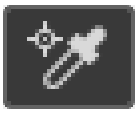
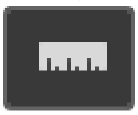
  - 
  - **Artboard Tool (V)**
  - The Artboard Tool allows you to easily design multiple web or UX (user experience) layouts for different devices or screen sizes.
  - 
  - **Rectangular Marquee Tool * (M)**
  - The Rectangular Marquee Tool draws rectangular selection outlines. Press and hold Shift as you drag to draw a square selection.
  - 

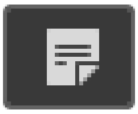
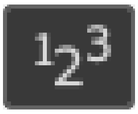
- **Elliptical Marquee Tool (M)**
- The Elliptical Marquee Tool draws elliptical selection outlines. Press and hold Shift to draw a selection in a perfect circle.
- 
- **Single Row Marquee Tool**
- The Single Row Marquee Tool in Photoshop selects a single row of pixels in the image from left to right.
- 
- **Single Column Marquee Tool**
- Use the Single Column Marquee Tool to select a single column of pixels from top to bottom.
- 
- **Lasso Tool * (L)**
- With the Lasso Tool, you can draw a freeform selection outline around an object.
- 
- **Polygonal Lasso Tool (L)**
- Click around an object with the Polygonal Lasso Tool to surround it with a polygonal, straight-edged selection outline.
- 
- **Magnetic Lasso Tool (L)**
- The Magnetic Lasso Tool snaps the selection outline to the edges of the object as you move your mouse cursor around it.
- 
- **Object Selection Tool * (W)**
- The Object Selection Tool lets you select an object just by dragging a rough selection outline around it.
- 
- **Quick Selection Tool (W)**
- The Quick Selection Tool lets you easily select an object simply by painting over it with a brush. Enable "Auto-Enhance" in the Options Bar for better quality selections.
- 
- **Magic Wand Tool (W)**
- Photoshop's Magic Wand Tool selects areas of similar color with a single click. The "Tolerance" value in the Options Bar sets the range of colors that will be selected.

Crop and Slice tools

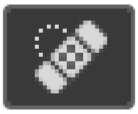
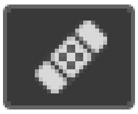
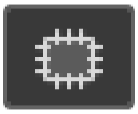
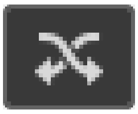
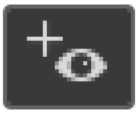
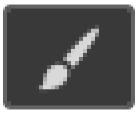
- 
- **Crop Tool * (C)**
- Use the Crop Tool in Photoshop to crop an image and remove unwanted areas. Uncheck "Delete Cropped Pixels" in the Options Bar to crop an image non-destructively.
- 
- **Perspective Crop Tool (C)**
- Use the Perspective Crop Tool to both crop an image and fix common distortion or perspective problems.
- 
- **Slice Tool (C)**
- The Slice Tool divides an image or layout into smaller sections (slices) which can be exported and optimized separately.
- 
- **Slice Select Tool (C)**
- Use the Slice Select Tool to select individual slices created with the Slice Tool.
- 
- **Frame Tool * (K)**
- New as of Photoshop CC 2019, the Frame Tool lets you place images into rectangular or elliptical shapes.

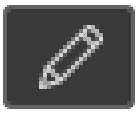
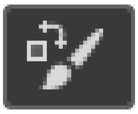
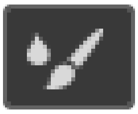
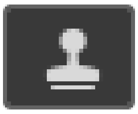
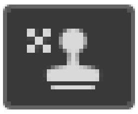
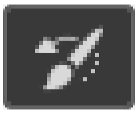
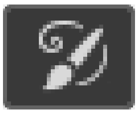
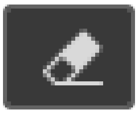
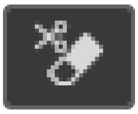
Measurement tools

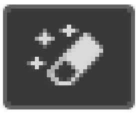
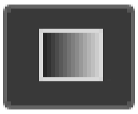
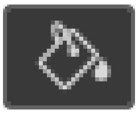
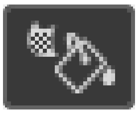
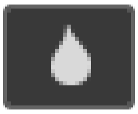
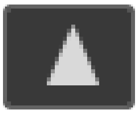
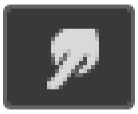
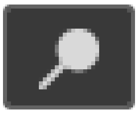
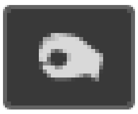
- 
- **Eyedropper Tool * (I)**
- Photoshop's Eyedropper Tool samples colors in an image. Increase "Sample Size" in the Options Bar for a better representation of the sampled area's color.
- 
- **3D Material Eyedropper Tool (I)**
- Use the 3D Material Eyedropper Tool to sample material from a 3D model in Photoshop.
- 
- **Color Sampler Tool (I)**
- The Color Sampler Tool displays color values for the selected (sampled) area in an image. Up to four areas can be sampled at a time. View the color information in Photoshop's Info panel.
- 
- **Ruler Tool (I)**

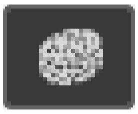
- The Ruler Tool measures distances, locations and angles. Great for positioning images and elements exactly where you want them.
- 
- **Note Tool (I)**
- The Note Tool allows you to attach text-based notes to your Photoshop document, either for yourself or for others working on the same project. Notes are saved as part of the .PSD file.
- 
- **Count Tool (I)**
- Use the Count Tool to manually count the number of objects in an image, or to have Photoshop automatically count multiple selected areas in the image.

Retouching and Painting tools

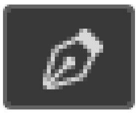
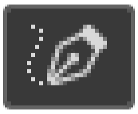
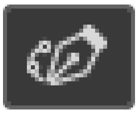
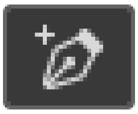
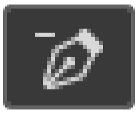
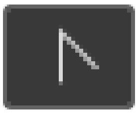
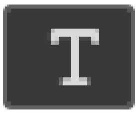
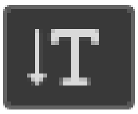
- 
- **Spot Healing Brush Tool * (J)**
- The Spot Healing Brush in Photoshop quickly removes blemishes and other minor problem areas in an image. Use a brush size slightly larger than the blemish for best results.
- 
- **Healing Brush Tool (J)**
- The Healing Brush lets you repair larger problem areas in an image by painting over them. Hold Alt (Win) / Option (Mac) and click to sample good texture, then paint over the problem area to repair it.
- 
- **Patch Tool (J)**
- With the Patch Tool, draw a freeform selection outline around a problem area. Then repair it by dragging the selection outline over an area of good texture.
- 
- **Content-Aware Move Tool (J)**
- Use the Content-Aware Move Tool to select and move part of an image to a different area. Photoshop automatically fills in the hole in the original spot using elements from the surrounding areas.
- 
- **Red Eye Tool (J)**
- The Red Eye Tool removes common red eye problems in a photo resulting from camera flash.
- 
- **Brush Tool * (B)**
- The Brush Tool is Photoshop's primary painting tool. Use it to paint brush strokes on a layer or on a layer mask.

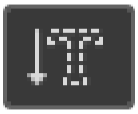
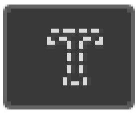
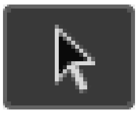
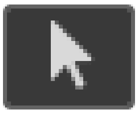
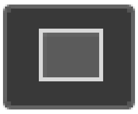
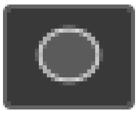
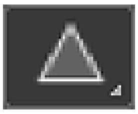
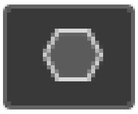
- 
- **Pencil Tool (B)**
- The Pencil Tool is another of Photoshop's painting tools. But while the Brush Tool can paint soft-edge brush strokes, the Pencil Tool always paints with hard edges.
- 
- **Color Replacement Tool (B)**
- Use the Color Replacement Tool in Photoshop to easily replace the color of an object with a different color.
- 
- **Mixer Brush Tool (B)**
- Unlike the standard Brush Tool, the Mixer Brush in Photoshop can simulate elements of real painting such as mixing and combining colors, and paint wetness.
- 
- **Clone Stamp Tool * (S)**
- The Clone Stamp Tool is the most basic of Photoshop's retouching tools. It samples pixels from one area of the image and paints them over pixels in another area.
- 
- **Pattern Stamp Tool (S)**
- Use the Pattern Stamp Tool to paint a pattern over the image.
- 
- **History Brush Tool * (Y)**
- The History Brush Tool paints a snapshot from an earlier step (history state) into the current version of the image. Choose the previous state from the History panel.
- 
- **Art History Brush Tool (Y)**
- The Art History Brush also paints a snapshot from an earlier history state into the image, but does so using stylized brush strokes.
- 
- **Eraser Tool * (E)**
- The Eraser Tool in Photoshop permanently erases pixels on a layer. It can also be used to paint in a previous history state.
- 
- **Background Eraser Tool (E)**

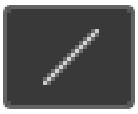
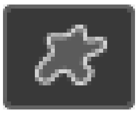
- The Background Eraser Tool erases areas of similar color in an image by painting over them.
- 
- **Magic Eraser Tool (E)**
- The Magic Eraser Tool is similar to the Magic Wand Tool in that it selects areas of similar color with a single click. But the Magic Eraser Tool then permanently deletes those areas.
- 
- **Gradient Tool * (G)**
- Photoshop's Gradient Tool draws gradual blends between multiple colors. The Gradient Editor lets you create and customize your own gradients.
- 
- **Paint Bucket Tool (G)**
- The Paint Bucket Tool fills an area of similar color with your Foreground color or a pattern. The "Tolerance" value determines the range of colors that will be affected around the area where you clicked.
- 
- **3D Material Drop Tool (G)**
- Used in 3D modeling, the 3D Material Drop Tool lets you sample a material from one area and then drop it into another area of your model, mesh or 3D layer.
- 
- **Blur Tool ***
- The Blur Tool blurs and softens areas you paint over with the tool.
- 
- **Sharpen Tool**
- The Sharpen Tool sharpens areas you paint over.
- 
- **Smudge Tool**
- The Smudge Tool in Photoshop smudges and smears the areas you paint over. It can also be used to create a finger painting effect.
- 
- **Dodge Tool * (O)**
- Paint over areas in the image with the Dodge Tool to lighten them.
- 
- **Burn Tool (O)**

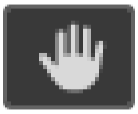
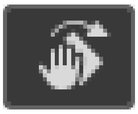
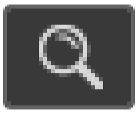
- The Burn Tool will darken the areas you paint over.
- 
- **Sponge Tool (O)**
- Paint over areas with the Sponge Tool to increase or decrease color saturation.

Drawing and Type tools

- 
- **Pen Tool * (P)**
- Photoshop's Pen Tool allows you to draw extremely precise paths, vector shapes or selections.
- 
- **Freeform Pen Tool (P)**
- The Freeform Pen Tool allows you to draw freehand paths or shapes. Anchor points are automatically added to the path as you draw.
- 
- **Curvature Pen Tool (P)**
- The Curvature Pen Tool is an easier, simplified version of the Pen Tool. New as of Photoshop CC 2018.
- 
- **Add Anchor Point Tool**
- Use the Add Anchor Point Tool to add additional anchor points along a path.
- 
- **Delete Anchor Point Tool**
- Click on an existing anchor point along a path with the Delete Anchor Point Tool to remove the point.
- 
- **Convert Point Tool**
- On a path, click on a smooth anchor point with the Convert Point Tool to convert it to a corner point. Click a corner point to convert it to a smooth point.
- 
- **Horizontal Type Tool * (T)**
- Known simply as the Type Tool in Photoshop, use the Horizontal Type Tool to add standard type to your document.
- 
- **Vertical Type Tool (T)**

- The Vertical Type Tool adds type vertically from top to bottom.
- 
- **Vertical Type Mask Tool (T)**
- Rather than adding editable text to your document, the Vertical Type Mask Tool creates a selection outline in the shape of vertical type.
- 
- **Horizontal Type Mask Tool (T)**
- Like the Vertical Mask Type Tool, the Horizontal Type Mask Tool creates a selection outline in the shape of type. However, the type is added horizontally rather than vertically.
- 
- **Path Selection Tool * (A)**
- Use the Path Selection Tool (the black arrow) in Photoshop to select and move an entire path at once.
- 
- **Direct Selection Tool (A)**
- Use the Direct Selection Tool (the white arrow) to select and move an individual path segment, anchor point or direction handle.
- 
- **Rectangle Tool * (U)**
- The Rectangle Tool draws rectangular vector shapes, paths or pixel shapes, with sharp or rounded corners. Press and hold Shift as you drag to force the shape into a perfect square.
- 
- **Ellipse Tool (U)**
- The Ellipse Tool draws elliptical vector shapes, paths or pixel shapes. Press and hold Shift as you drag to draw a perfect circle.
- 
- **Triangle Tool (U)**
- The Triangle Tool draws triangle shapes. Hold Shift to draw an equilateral triangle, or use the Radius option to round the corners.
- 
- **Polygon Tool (U)**
- The Polygon Tool draws polygonal shapes with any number of sides. Use the Star Ratio option to turn polygons into stars.

- 
- **Line Tool (U)**
- The Line Tool draws straight lines or arrows. Use the Stroke color and weight to control the appearance of the line.
- 
- **Custom Shape Tool (U)**
- Photoshop's Custom Shape Tool lets you select and draw custom shapes. Choose from Photoshop's hundreds of built-in custom shapes or create your own.

- ## Navigation tools
  - 
  - **Hand Tool * (H)**
  - The Hand Tool lets us click and drag an image around on the screen to view different areas when zoomed in.
  - 
  - **Rotate View Tool (R)**
  - Use the Rotate View Tool in Photoshop to rotate the canvas so you can view and edit the image from different angles.
  - 
  - **Zoom Tool * (Z)**
  - Click on the image with the Zoom Tool to zoom in on a specific area. Press and hold Alt (Win) / Option (Mac) and click with the Zoom Tool to zoom out.

Resolution and Image Size

Resolution refers to the number of pixels in an image per inch. Resolution is sometimes identified by the width and height of the image as well as the total number of pixels in the image. For example, an image that is 2048 pixels wide and 1536 pixels high (2048X1536) contains (multiply) 3,145,728 pixels (or 3.1 Megapixels). You could call it a 2048X1536 or a 3.1 megapixel image.

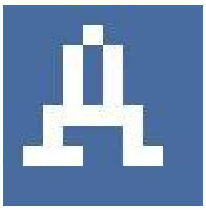
Resolution is typically described in PPI, which refers to how many pixels are displayed per inch of an image. Higher resolutions mean that there more pixels per inch (PPI), resulting in more pixel information and creating a high-quality, crisp image.

Images with lower resolutions have fewer pixels, and if those few pixels are too large (usually when an image is stretched), they can become visible like the image below.

When you change the resolution of an image, you are saying how many pixels you want to live in each inch of the image. For example, an image that has a resolution of 600 ppi will contain 600 pixels within each in of the image. 600 is a lot of pixels to live in just one inch, which is why 600ppi images will look very crisp and detailed. Now, compare that to an image with 72ppi, which has a lot fewer pixels per inch.

Let we take good example; let's create an image of the letter "a," going from the lowest possible resolution to something much larger than that. At a resolution of 1 by 1, (a square that's one dot by one dot), we end up with an "image" that looks like this:

Well that's pretty boring. Why no letter A? There is no letter A because there are no more dots than the single dot we used. So let's bump up the resolution and go with a 10 by 10 square.



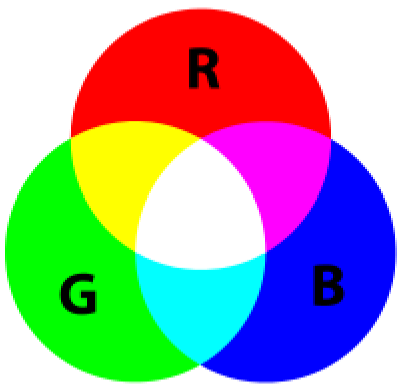
Now, it looks like an A. Hopefully you can see very clear here. The more we increase the resolution of the graphic, the clearer the A becomes. Now let's multiply the 10 by 10 resolution by 5 to get a 50 by 50 square.

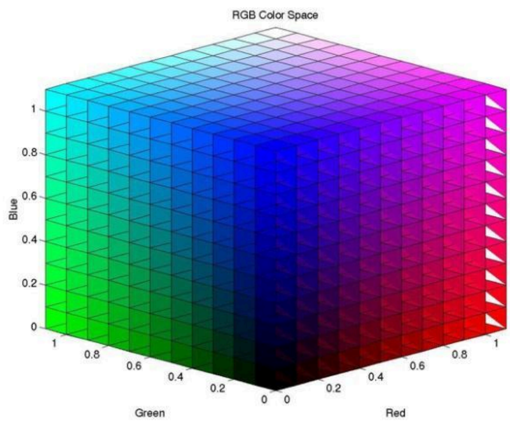
Color Spaces and Color Modes

Color space, also known as the color model (or color system), is an abstract mathematical model which simply describes the range of colors as tuples of numbers, typically as 3 or 4 values or color components (e.g. RGB or CMYK).

A color space is a useful method for users to understand the color capabilities of a particular digital device or file. It represents what a camera can see, a monitor can display or a printer can print, and etc. There are a variety of color spaces, such as RGB, CMY, HSV, HIS. We will talk about most popular RGB and CMYK color spaces.

RGB





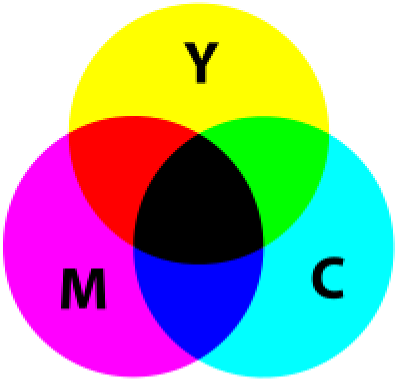
RGB (R=Red, G=Green, B=Blue) is a kind of color space which uses red, green and blue to elaborate color model. An RGB color space can be simply interpreted as "all possible colors" which can be made from three colors for red, green and blue. In such conception, each pixel of an image is assigned a range of 0 to 255 intensity values of RGB components. That is to say, using only these three colors, there can be 16,777,216 colors on the screen by different mixing ratios.

The variant of RBG are sRBG (standard RGB) was developed by HP and Microsoft. And Adobe RGB is developed by Abobe. Generally, RGB is used in Monitor, digital device screen, Camera, Camera print and small home use printers.

This is additive color models, white is the "additive" combination of all primary colored lights, while black is the absence of light.

CMYK

CMYK (C=Cyan, M=Magenta, Y=Yellow and K=Key or Black). The CMYK color model is a subtractive color model, used in color printing, and is also used to describe the printing process itself. CMYK refers to the four inks used in some color printing: cyan, magenta,

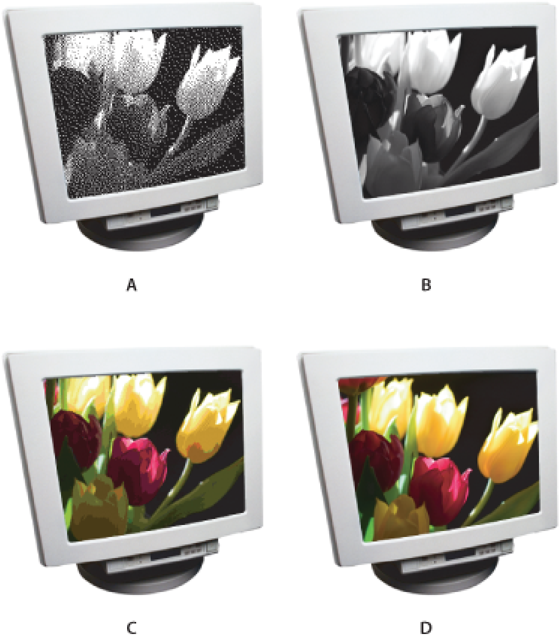


yellow, and key.

The press operator, press manufacturer, and press run, ink is typically applied in the order of the abbreviation. The "K" in CMYK stands for key because in four-color printing, cyan, magenta, and yellow printing plates are carefully keyed, or aligned, with the key of the black key plate. Such a model is called subtractive because inks "subtract" brightness from white.

Color Modes

An image mode determines the number of colors that can be displayed in an image. It can also affect the file size of the image. Photoshop Elements provides four image modes: RGB, bitmap, grayscale, and indexed color.



A. Bitmap mode B. Grayscale mode C. Indexed-color mode D. RGB mode

Bitmap mode

It uses one of two color values (black or white) to represent the pixels in an image. Images in bitmap mode are called 1-bit images because they have a bit depth of 1.

Grayscale mode

It uses up to 256 shades of gray. Grayscale images are 8-bit images. Every pixel in a grayscale image has a brightness value ranging from 0 (black) to 255 (white). Grayscale values can also be measured as percentages of black ink coverage (0% is equal to white, 100% to black).

Indexed Color mode

It uses up to 256 colors. Indexed-color images are 8-bit images. Indexed color can reduce file size while maintaining visual quality—for example, for a web page. Limited editing is available in this mode. For extensive editing, you should convert temporarily into RGB mode.

RGB Color mode

It is the default mode of new Photoshop images and images from your digital camera. In RGB mode, the red, green, and blue components are each assigned an intensity value for every pixel—ranging from 0 (black) to 255 (white). For example, a bright red color might have an R value of 246, a G value of 20, and a B value of 50. When the values of all three components are equal, the result is a shade of neutral gray. When the value of all components is 255, the result is pure white; when the value is 0, the result is pure black.

Selection tools, Filling stroking tools

Selections, Filling, and Stroking are fundamental concepts in Photopea (and similar programs like Photoshop) that let you precisely control where you apply effects or color.

Here is an explanation of the tools and how they work together, along with examples:

Selection Tools in Photopea

Selection tools define a specific area of your image that you want to isolate for editing. Only the selected area will be affected by subsequent operations (like filling or stroking).

Tool	Purpose	Example Use
Rectangle/Ellipse Select (Marquee Tools)	Selects a rectangular or elliptical area.	Use the **Rectangle Select Tool** to select a **perfect square** (hold **Shift** while dragging) for a frame or a **circle** (using **Ellipse Select** with **Shift**) to select a ball.
Lasso Tools (Lasso, Polygonal, Magnetic)	For free-form selections.	* **Lasso Tool:** Freehand drawing a very rough selection around a cloud.
Magic Wand	Selects pixels of a similar color in a single click.	Clicking on a **blue sky** to quickly select all the similar shades of blue, making it easy to change only the sky's color.
Quick Selection Tool	Intelligently selects complex areas by detecting edges.	Dragging over a **person's outline** in a photo. The tool tries to "stick" to the edges of the person, allowing you to select them to cut them out or change the background.

Filling and Stroking a Selection

Once you have an active selection (indicated by a moving, dashed line, often called "marching ants"), you can use the **Fill** or **Stroke** commands to modify *only* that area.

1. Filling a Selection (Coloring the Inside)
Filling replaces all the pixels *inside* the selection with a specified color, pattern, or content.

Method	Steps	Example
Foreground Color	Press **Alt + Delete** (Windows) or **Option + Delete** (Mac).	You've selected a broken section of a road sign. Pressing **Alt + Delete** immediately fills the selected area with the foreground color (e.g., bright yellow) to "fix" it.
Background Color	Press **Ctrl + Delete** (Windows) or **Cmd + Delete** (Mac).	You've selected a shape you want to fill with pure white. Set the background color to white and press the shortcut.
Using the Fill Menu	Go to **Edit > Fill** and choose your content (Foreground, Background, Pattern, Content-Aware, etc.).	You select an object on a busy background, then use **Edit > Fill > Content-Aware** to intelligently remove the object and fill the gap with surrounding pixels.

2. Stroking a Selection (Adding an Outline)
Stroking creates a line (or outline) along the boundary of the selection.

Method	Steps	Example
Stroke Command	Go to **Edit > Stroke**. A dialog box appears where you set the width, color, and location (Inside, Center, or Outside) of the stroke.	You create a rectangular selection on a photo. Using **Edit > Stroke** with a **10px width**, **Red** color, and **Outside** position adds a crisp, red 10-pixel border around the area.

Blending options

Blending Options it means The Magical Effects

This is where your decoration gets fancy! Blending Options are like putting special **magical stickers** on your shapes that change how they look without ruining the original shape.

1. Blending Modes (Layer Blending)

This is how one sheet of paper (Layer 1) changes the color of the paper beneath it (Layer 2).

- **Normal:** They just stack up. Layer 1 completely hides Layer 2 where it covers it.

- **Multiply (The Shadow Maker):** This mode makes colors **darker**, like shining a shadow on the paper below. If you put a gray circle on a yellow background and set it to Multiply, the yellow turns into a **darker, golden yellow**.
- **Screen (The Light/Glow Maker):** This mode makes colors **lighter**, like shining a glow stick on the paper below. If you put a white shape on a photo and set it to Screen, it makes a **beautiful bright flare** or glow.
- **Color:** This changes the **color** of the paper underneath, but it keeps the **lightness and darkness** exactly the same. Good for quickly changing a blue shirt to a red shirt without losing the folds and wrinkles.

2. Layer Styles (The Magic Stickers)

These are effects that *stick* to your shape and follow it everywhere without ever being permanent.

Layer Style	**Analogous to...**	**What it Does**
Drop Shadow	Putting a **coin** on your desk.	It makes the shape look like it's **lifting up** off the paper, casting a shadow underneath.
Bevel & Emboss	**Carving** the shape out of wood or stone.	It makes the edges look **round and shiny** (raised) or **sunken in** (carved), giving it a 3D feel.
Gradient Overlay	Covering the shape with **rainbow paint**.	It colors the whole shape not just one color, but a **smooth transition** from blue to green to yellow.

Content Adjustment option

Content Adjustment" in Photopea refers to a powerful set of tools designed to seamlessly remove objects, repair imperfections, and intelligently move content within an image.

These tools are often grouped under the **"Healing Tools"** and the **"Fill"** command.

🩹 Healing and Cloning Tools

These tools allow you to replace unwanted content by sampling (copying) pixels from another area and blending them to match the new surroundings.

Tool Name	Icon	Function / Best Use	Practical Example
Spot Healing Brush Tool		**Automatic removal** of small flaws. You just click or paint over the problem area. Photopea intelligently analyzes the surrounding pixels to replace the content.	Removing a **pimple or dust spot** from a photo. The tool automatically fills the spot with blended skin or background texture.
Healing Brush Tool		Similar to the Spot Healing Brush, but **you manually define the source**. You **Alt/Option-click** to sample a clean area, then paint over the flawed area.	Repairing a **crack in a wall**. You sample a clean section of the wall near the crack and then paint over the crack. The tool copies the texture but *blends* the color and tone to match the flawed area, making the repair seamless.
Clone Stamp Tool		**Directly copies** pixels from a source area to a destination area. No color/tone blending is applied automatically.	Duplicating an object (like a flower). You **Alt/Option-click** on the flower, then paint elsewhere. It's also used for **touch-up blending** when other tools create repeating patterns.
Patch Tool		Best for **replacing large, irregular areas**. You draw a selection around the area you want to replace (the flaw), then drag the selection to a clean area to sample.	Removing a **large stain** from a uniform background. The sampled area is used to replace the stain, and it automatically blends the edges.
Content-Aware Move Tool		Allows you to **move** an object and have the empty gap filled intelligently.	Selecting a **person** and moving them to a different spot in the frame. The tool moves the person and uses surrounding pixels to fill the original spot they left behind.

🗑️ Content-Aware Fill

The **Content-Aware Fill** is not a tool on the toolbar but a powerful command found in the menu, designed to remove large objects entirely.

How it Works

1. **Select the Object:** Use any selection tool (like the **Lasso Tool**) to draw a selection tightly around the unwanted object (e.g., a car in a field).
2. **Access the Command:** Go to **Edit > Fill** (or press **Shift + F5**).

3. **Choose the Content:** In the dialog box, set the **Fill** dropdown to **Content-Aware**.
4. **Execute:** Click **OK**.

Practical Example:

Imagine you have a beautiful landscape photo, but there is one **small signpost** you want to remove.

- You select the signpost using the **Lasso Tool**.
- You choose **Edit > Fill > Content-Aware**.

Photopea then analyzes the surrounding environment (grass, sky, trees) and uses that information to **intelligently generate new pixels** to fill the hole left by the signpost, making it look as if the signpost was never there.

History option.

Key Concepts of the History Panel

1. History States

Every action you take (a brush stroke, a layer move, applying a filter, cropping the image) is recorded as a **History State** in the panel.

- **Access:** If the panel isn't visible, you can open it by going to **Window > History**.
- **The List:** The panel displays a chronological list of these states, with the oldest action at the top and the most recent one at the bottom.
- **Time Travel:** By **clicking on any state** in the list, you instantly revert your image to how it looked right after that action was completed. This is much faster than repeatedly pressing "Undo."

2. How to "Undo" and "Redo"

While the History Panel is the visual way to go back, the quickest way to step through history is using keyboard shortcuts:

Action	Menu Command	Keyboard Shortcut (Windows)	Keyboard Shortcut (Mac)
Undo / Toggle Last Action	Edit > Undo	**Ctrl + Z**	**Cmd + Z**
Step Backward (Multi-Undo)	Edit > Step Backward	**Ctrl + Alt + Z**	**Cmd + Option + Z**
Step Forward (Redo)	Edit > Step Forward	**Ctrl + Shift + Z**	**Cmd + Shift + Z**